

GreenRisk: An Explainable-by-Construction Fuzzy Instrument for Scoring Greenwashing Risk in Corporate Climate Disclosures

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Abstract

Corporate climate disclosures are increasingly common, but their volume makes it difficult to distinguish substantive commitments from well-worded aspirations. Measuring this difference remains challenging because no large-scale ground-truth corpus of greenwashing exists. Existing approaches often examine a single linguistic dimension or combine several signals arithmetically, obscuring their interaction. This work presents GreenRisk, an explainable fuzzy-inference instrument that combines four construct-aligned language signals—specificity, commitment, sentiment asymmetry, and net-zero targeting—through a human-readable Mamdani rule base. For each paragraph, it produces a risk score on a 0–100 scale and a traceable path through the rules that shaped the score. The instrument was evaluated using a pre-registered, two-layer protocol designed for a setting without a large-scale ground-truth corpus. On 1,009 TCFD disclosure paragraphs, GreenRisk correlates with an established cheap-talk baseline (Spearman $\rho = 0.602$) and continues to differentiate paragraphs within the baseline’s vaguest quartile ($\rho = 0.433$, $n = 253$). On a held-out case set, it distinguishes five regulator-adjudicated greenwashing cases from six rigorous reference cases with an AUC of 0.867. These results suggest that combining linguistic signals adds information beyond vagueness alone while keeping the reasoning open to inspection. GreenRisk and its evaluation artifacts are released under an open license and are intended as a decision-support layer for human auditors.

CCS Concepts

• **Computing methodologies** → **Vagueness and fuzzy logic**; *Natural language processing*; • **Applied computing** → *Environmental sciences*.

Keywords

esg, greenwashing, fuzzy inference, explainability, climate disclosures, provenance, construct validity

1 Introduction

Corporate climate disclosure has become a widely used instrument, produced at industrial scale by sustainability teams, communications departments, and legal counsel. These disclosures are consumed by regulators, investors, and the public, who have little practical means of verifying them paragraph by paragraph. At this volume, distinguishing a substantive climate commitment from a

well-worded aspiration becomes both a trust problem and a measurement problem. A 2021 European Commission screening of online green claims found that 42% were exaggerated, false, or otherwise deceptive [2], while a 2022 Google Cloud and Harris Poll survey found that 58% of corporate executives admitted that their own organizations engage in greenwashing [3]. Together, these findings describe a disclosure ecosystem in which vagueness, exaggeration, and selective emphasis are common.

A growing body of natural language processing research addresses parts of this problem, most notably through the ClimateBERT family of classifiers, an open suite of transformer models fine-tuned on climate-specific corpora for tasks such as topic detection, specificity, commitment, and sentiment framing [1, 6, 9]. These models provide validated, single-purpose assessments of individual linguistic constructs. The composition of these constructs, however, remains underexplored, since a paragraph may express several constructs at once. Existing greenwashing-risk instruments tend either to score a single dimension in isolation, such as vagueness, or to combine several signals through simple arithmetic, such as an unweighted average, without an explicit and inspectable account of how the result was reached. For example, high vagueness and high commitment have a different risk meaning together than either has alone, yet an arithmetic average obscures this interaction.

This gap motivates the main research question: can greenwashing risk be decomposed into a small set of construct-aligned linguistic signals and recombined through an explicit, human-readable rule base into a single auditable score that is explainable by design?

To address this question, this paper makes five contributions. First, it presents a method for designing a linguistically decomposed risk instrument built from multiple version-pinned, construct-aligned signals. Second, it releases the resulting artifact, GreenRisk, including its locked rule base, membership functions, and model registry, archived with a Zenodo DOI for reproducibility. Third, it proposes an evaluation protocol suited to the absence of large-scale greenwashing ground truth: the instrument is locked, the evaluation criteria are pre-registered, and a held-out case set is scored once. Fourth, it treats provenance as a first-class deliverable by binding every scored run to its exact model revisions and instrument version through a W3C PROV-O graph. Fifth, it reports empirical findings from applying GreenRisk to a large TCFD corpus and a held-out, regulator-adjudicated case set, together with the limitations those findings reveal.

The remainder of this paper presents the background on which the design draws (Section 2), the instrument itself (Section 3), the

evaluation design and its rationale (Section 4), the corpus- and case-level results (Section 5), the instrument's limitations (Section 6), and the conclusions and directions for future work (Section 7).

2 Background and Related Work

Measuring greenwashing lies at the intersection of three fields that are often treated separately: climate-disclosure regulation, natural language processing, and fuzzy control theory. Each supplies a piece of the design this paper develops and helps explain the choices underlying it.

2.1 The ClimateBERT family

At the center of the language layer is ClimateBERT, a transformer-based language model further pretrained on a large corpus of climate-related text and released as part of an open, Apache-2.0-licensed model family [9]. Building on this foundation, Bingler et al. fine-tune and validate classifiers for climate-specific constructs, including a specificity classifier that distinguishes concrete, figure-bearing text from vague language. They also introduce the “cheap talk” measure—one minus the specificity model’s probability of the specific class—as a validated proxy for rhetorical vagueness in climate disclosures [1]. Schimanski et al. extend the family with a three-way classifier that distinguishes among the absence of a target, a reduction target, and a net-zero target. This distinction matters because net-zero pledges carry a different evidentiary burden from quantified reduction goals [6]. Together, these models provide GreenRisk with four signals tied to validated, independently published constructs.

2.2 Mamdani fuzzy inference

Central to the reasoning layer is Mamdani fuzzy inference, a rule-based inference model introduced by Mamdani and Assilian for linguistic control problems in which precise numeric thresholds are less natural than graded, human-readable terms such as low, medium, and high [5]. A Mamdani system represents each input as a degree of membership in overlapping linguistic terms. It combines those degrees through rules such as “if specificity is low and commitment is high, then risk is high” and produces a numeric output through defuzzification, typically using the centroid of the aggregated output membership. Its rule-level transparency has made Mamdani inference useful in control engineering for decades: each rule can be read and evaluated independently. GreenRisk uses the same property for text scoring.

2.3 Explainability by construction

Explainability in machine learning is most often sought after the fact, through methods such as feature attribution or surrogate modeling that approximate why an opaque model produced a particular output. GreenRisk instead pursues explainability by construction: every intermediate quantity, from the raw signal probabilities and linguistic-term memberships to the specific rules that fired, is inspectable. The explanation is therefore not a separate artifact computed after scoring but an integral part of the scoring process. This distinction is especially important for greenwashing risk, where the score itself constitutes a claim subject to examination.

2.4 Provenance as a first-class artifact

Beyond explaining a single score, reproducing it across time requires provenance, and the W3C PROV-O ontology supplies a standard, machine-readable vocabulary for recording which agents, activities, and entities produced a given result [4]. GreenRisk uses PROV-O to bind each scored run to the exact pinned model revisions, the input paragraphs, and the instrument version tag that produced it, so that a claim about a score’s provenance is a graph that can be checked.

2.5 Regulatory context

The measurement problem also sits within a changing regulatory context. The Task Force on Climate-related Financial Disclosures (TCFD) established a voluntary disclosure framework organized around governance, strategy, risk management, and metrics and targets, and remains the source of the corpus used for this paper’s large-scale evaluation [8]. The European Union’s Corporate Sustainability Reporting Directive (CSRD) has since moved much of this territory from voluntary to mandatory disclosure, a shift that raises the stakes of measuring disclosure quality correctly without resolving the underlying vagueness problem on its own.

3 The GreenRisk Instrument

At the core of GreenRisk is a five-stage pipeline that turns one paragraph of climate disclosure into one auditable score, with a provenance graph running alongside the run:

- (1) **Climate-relevance gate** — is this paragraph about climate at all?
- (2) **Four signals** — four independent, pinned ClimateBERT reads.
- (3) **Fuzzification** — four probabilities translated into linguistic-variable membership degrees.
- (4) **Rule base** — a 17-rule Mamdani inference, with rule combination by AND = min.
- (5) **Defuzzification** — centroid defuzzification into a risk score in [0, 100], alongside a fired-rule trace.

A parallel provenance stage wraps every run in a W3C PROV-O graph, binding the result to the exact pinned model revisions and instrument version that produced it.

3.1 Stage 1: the climate gate

Not every paragraph in a disclosure document is about climate, and scoring an unrelated paragraph for greenwashing risk would produce a number with no referent. GreenRisk therefore begins with a fifth, independent classifier from the ClimateBERT family, pinned in the model registry. It answers a binary question—whether the paragraph is climate-relevant—and gates on $P(\text{‘yes’}) \geq 0.5$. Paragraphs that fail the gate are dropped before they reach the fuzzy layer, and the rule base itself assumes this gating has already happened.

3.2 Stage 2: the four signals

Each signal, summarized in Table 1, is read from an independently pinned ClimateBERT classifier. Every signal is construct-aligned: a

Table 1: The four risk signals, each read from an independently pinned ClimateBERT classifier. Inversion to a risk direction happens later, in the rule base, never in the signal itself.

Signal	Reads	What it measures
specificity	$P(\text{spec})$	Concrete figures, dates, or mechanisms, versus vague language
commitment	$P(\text{yes})$	A pledge or claim, versus a statement of fact
sentiment_ asymmetry	$P(\text{opportunity})$	Opportunity-framed text, versus risk-framed or neutral text
netzero	$P(\text{net-zero})$	A net-zero target specifically, distinct from a general reduction target

high specificity score, for example, means that the text is specific. Any inversion to a risk direction occurs later in the rule base.

Two mapping choices warrant explanation because the underlying models permit other interpretations. First, *netzero* reads $P(\text{net-zero})$ alone. It excludes $P(\text{reduction})$. The underlying model is a three-way classifier (none / reduction / net-zero); reduction targets are quantified against a base year and are conceptually the opposite of the vagueness this instrument targets, so folding them in would dilute the specific “net-zero pledge with low specificity” signature the project is built to catch. Second, *sentiment_asymmetry* reads $P(\text{opportunity})$ alone; it does not calculate a signed combination of opportunity and risk framing. Consequently, risk-framed and neutral paragraphs both collapse toward zero on this signal, so “balanced” framing does not sit at a distinguishable 0.5. This deliberate simplification affects how the opportunity-framing rules (Tier 3, below) should be interpreted.

3.3 Membership functions

All four inputs share the same triangular geometry on $[0, 1]$: Low is $(0, 0, 0.4)$, Medium is $(0.2, 0.5, 0.8)$, and High is $(0.6, 1, 1)$. The output risk, on $[0, 100]$, uses four triangular terms for finer resolution: Low $(0, 0, 30)$, Moderate $(15, 35, 55)$, Elevated $(45, 65, 85)$, and High $(70, 100, 100)$.

The 0.4/0.6 breakpoints, shown for the specificity input in Figure 1, are a theory-driven default, symmetric around the 0.5 calibration midpoint of the underlying binary classifiers. They were not fitted to a particular corpus. A corpus-quantile boundary would be circular: it would define “vague” relative to the sample instead of in absolute terms. Rule combination uses $\text{AND} = \min$, and centroid defuzzification over the aggregated output membership produces the final score.

3.4 Stages 3–4: the 17-rule base

The central design principle is that risk equals vagueness multiplied by claim-strength. A vague paragraph that makes no claim is a weak disclosure, not greenwashing; it simply has nothing to evaluate. A vague paragraph that makes a confident, unspecific claim is the classic greenwashing signature, the promise without

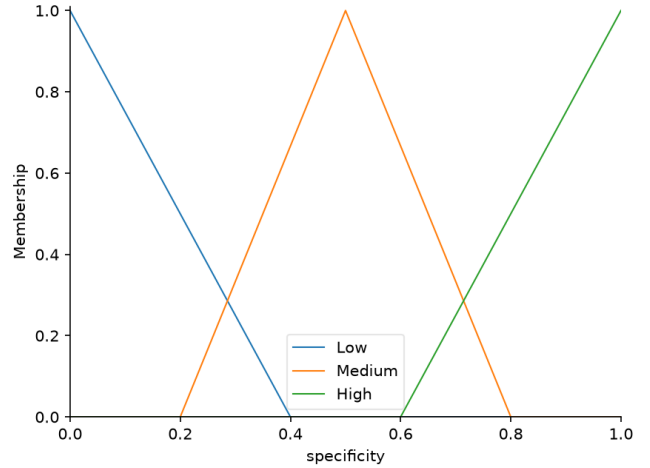


Figure 1: The locked input membership functions on the specificity signal (Low/Medium/High, triangular). The trapezoidal alternative tested under DL-002 is overlaid for comparison but is not part of the scored path.

the substance behind it. The rule base is built so that risk is monotone non-increasing in specificity, and monotone non-decreasing in commitment specifically when specificity is low, so that a loud, unsubstantiated pledge is scored higher than either a quiet non-claim or a specific, substantiated one.

Table 2 lists all 17 rules across four tiers. Tier 1, the spine, covers all nine combinations of specificity and commitment. Tier 2 escalates or exonerates specifically around net-zero claims, so that a net-zero pledge with no detail (N1) or no commitment language (N2) escalates, while a net-zero claim paired with high specificity (N3) is exonerated. Tier 3 escalates when strongly opportunity-framed text is also vague (O1) or uncommitted (O2), and does not escalate when opportunity framing is paired with both high specificity and high commitment (O3, “earned optimism”). Tier 4 defines two classic composite patterns: a vague, uncommitted net-zero pledge (G1) and a vague, optimistically-framed net-zero pledge (G2), kept separate from the spine so the audit trace can point at “this matched the classic greenwashing pattern” in case studies, in addition to the graded spine score.

3.4.1 DL-001: demoting the “absence” corner. The spine in Table 2 is not the rule base’s first version, and the correction that produced it is the design’s central discovery. The first traced inference run, on 21 climate-gated TCFD paragraphs, found that 14 of those 21 genuine climate paragraphs maxed out at risk ≈ 90 through rule S1 alone. Reading them showed they were sober climate risk-factor disclosures, such as “climate change could exacerbate threats to our business, including exposure to severe storms, drought, fires, and floods”, the type of reporting TCFD exists to encourage. The root cause was that S1’s original premise, that vague and uncommitted together mean cheap talk, is false for risk-disclosure text: such text is non-specific and uncommitted because it is not a pledge at all. Greenwashing is a hollow claim; risk-factor disclosures are not

Table 2: The locked 17-rule Mamdani base (rulebase-locked-v1). Abbreviations: *spec* = specificity, *comm* = commitment, *opp* = sentiment_asymmetry (opportunity framing), *nz* = netzero; L/M/H = Low/Medium/High membership term. AND = min combines multi-term antecedents.

Tier	Rule	Antecedent	Risk term
1 — spine (spec × comm)	S1	spec L, comm L	Moderate
	S2	spec L, comm M	Elevated
	S3	spec L, comm H	High
	S4	spec M, comm L	Moderate
	S5	spec M, comm M	Moderate
	S6	spec M, comm H	Elevated
	S7	spec H, comm L	Low
	S8	spec H, comm M	Low
	S9	spec H, comm H	Low
2 — net-zero amplifiers	N1	nz H, spec L	High
	N2	nz H, comm L	Elevated
	N3	nz H, spec H	Low
3 — opportunity-framing amplifiers	O1	opp H, spec L	Elevated
	O2	opp H, comm L	Elevated
	O3	opp H, spec H, comm H	Low
4 — named signatures	G1	spec L, comm L, nz H	High
	G2	spec L, opp H, nz H	High

claims, and the original rule base could not tell an empty pledge from no pledge whatsoever.

The fix operated at the rule level by requiring a claim signal for High risk: commitment High, netzero High, or opportunity High. Five spine cells changed. S1 moved from High to Moderate for a vague paragraph with no claim, while S3 moved from Elevated to High for a loud pledge with no substance. S4 moved from Elevated to Moderate for a partially specific non-claim. S6 moved from Moderate to Elevated for a claim with partial backing, and S7 moved from Moderate to Low for a specific factual disclosure. Measured on the same 21-paragraph sample, the number of paragraphs maxing out above 80 fell from 14/21 to 1/21, and mean risk fell from 77.3 to 36.9; the pattern held at scale on a 162-paragraph validation sample, where maxed-out cases fell to 1/162.

Two short paragraphs illustrate why the spine is shaped this way. The sentence “Climate change could exacerbate threats to our business, including exposure to severe storms, drought, fires, and floods” has low specificity, no figures, and low commitment because it is a factual risk statement, not a pledge. Rule S1 therefore returns Moderate: the paragraph is treated as a weak disclosure, not as greenwashing. The sentence “We are committed to a net-zero future for all” has the same low specificity but high commitment, forming a confident claim with nothing concrete behind it. Rule S3 returns High, the greenwashing signature. The corrected spine separates these cases through an inspectable rule; no tuned threshold is needed.

3.4.2 DL-002: membership-function shape, triangular kept. One open instrument decision remained before the lock: whether the specificity “Low” term should stay triangular, peaking only at specificity = 0, or become trapezoidal, with a flat plateau near zero that would treat the whole near-zero pile-up of vague text as fully Low. Both variants were run against the same 162-paragraph

validation sample, holding the rule base and the other three inputs fixed. The two variants produced identical per-band counts, near-identical mean risk (25.92 versus 25.94) and spread, and zero band-boundary crossings out of 162 paragraphs. The only movement anywhere was a single paragraph edging from 89.20 to 90.00, still within the High band. The triangular form was kept, since it uses fewer parameters for no measurable gain in discrimination; the trapezoidal variant remains in the codebase as a documented, reproducible comparison point outside the scoring path.

3.5 Stage 5: output, trace, and provenance

score_paragraph returns the final centroid score together with a trace of every rule that fired, sorted by firing strength and independently re-derived from the locked membership functions. The trace does not depend on the inference library’s internal state, allowing the two computations to be checked against each other. A parallel provenance module wraps each scoring run in a W3C PROV-O graph that binds the result to the pinned model revisions and instrument version tag. This graph records what produced each score.

3.6 The locked instrument

The rule base and membership functions described above are frozen under the annotated tag rulebase-locked-v1 (commit a40288a). After this point, any change to the rules, the membership functions, the model revisions, or the signal mappings is logged as a separate instrument version. This matters because all of the validation evidence in Section 5 was collected against this locked instrument, including a held-out test set that was not read or scored until after the freeze.

4 Evaluation Design

Central to the evaluation design is a constraint most classifier evaluations do not face: no corpus of greenwashing-labeled paragraphs exists at the scale or quality this project would need, and none of the companies in the held-out case set has ever been exhaustively labeled paragraph by paragraph. Reporting precision, recall, or F1 against invented or crowd-sourced labels would answer a different question: whether GreenRisk agrees with whoever produced those labels. It would not establish that the instrument detects greenwashing. This work therefore adopts construct validity as its evaluation frame and asks whether the score behaves as a valid measure of the construct should under known conditions.

This validity argument is built in two independent layers. The first layer establishes convergent and discriminant validity at corpus scale, comparing GreenRisk’s score against the Binger cheap-talk baseline, an already-validated single-signal measure, across 1,009 climate-gated TCFD paragraphs. The second layer establishes face validity at case-study scale, checking whether GreenRisk’s score separates real, regulator-adjudicated greenwashing cases from independently verified rigorous disclosures. The corpus layer is large but proxy-based because cheap talk is not ground truth. The case-study layer is closer to ground truth but necessarily small because regulator-adjudicated cases do not exist in bulk. Together, the layers triangulate validity using sources of evidence with different limitations.

Before any case in the held-out contrast set was scored, it was read and stratified in advance into four groups: in-scope vagueness-based greenwashing, a boundary promotional case, out-of-scope specificity-based greenwashing, and a rigorous reference group. This pre-registration step is what allows an out-of-scope miss to be interpreted afterward as a documented scope limit. The held-out case set was not inspected or scored until after the instrument was frozen under rulebase-locked-v1, and was then scored once. This one-shot discipline exists for the same reason a pre-registered clinical trial does not permit a second look: an instrument that can be adjusted after seeing the test set is being fit to that set, which invalidates it as an independent evaluation.

Layer 1 reports Spearman rank correlation because the score distribution does not satisfy the assumptions motivating Pearson’s linear correlation. The four well-separated output terms produce clusters near their centroids, roughly 11, 35, 65, and 90, instead of a continuous spread across $[0, 100]$. Rule S1 alone is the top-firing rule for 428 of the 1,009 gated paragraphs, creating heavy ties near the Moderate centroid of 35. Spearman correlation accommodates this clustered, tie-heavy distribution without imposing a linear relationship.

5 Results

The evidence gathered against the locked instrument spans two scales: a corpus of more than a thousand paragraphs and a curated set of fifteen cases. Each scale addresses a different part of the validity argument.

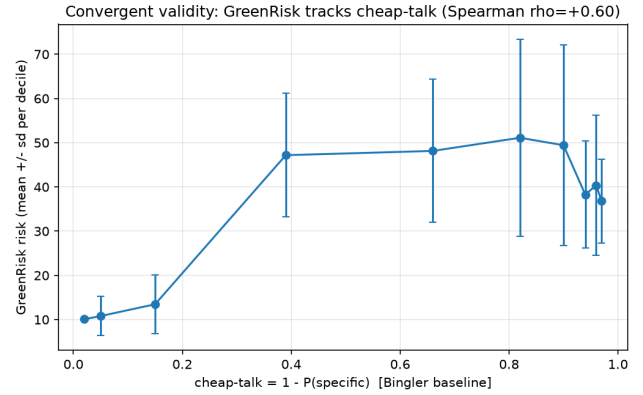


Figure 2: Mean GreenRisk score by decile of Binger cheap-talk. The curve climbs through the middle deciles but bends down in the vaguest decile, the DL-001 signature surfacing at corpus scale.

5.1 Layer 1: corpus-scale convergent and discriminant validity

Of the 1,300 paragraphs sampled from the `climatebert/tcfd_` recommendations corpus, 1,009 passed the climate-relevance gate and were scored by the full pipeline. The remaining 291 were dropped as not climate-relevant before reaching the fuzzy layer. Scored against the Binger et al. cheap-talk baseline, GreenRisk’s risk score correlates strongly and in the expected direction, Spearman $\rho = 0.602$ ($p \approx 1.4 \times 10^{-100}$, $n = 1,009$), establishing that the instrument tracks the same underlying vagueness signal as an already-validated external measure.

The relationship in Figure 2 is non-monotone at the extreme, reflecting the DL-001 rule design at corpus scale. Mean risk rises with cheap talk through the middle deciles but bends downward in the vaguest decile. In that decile, cheap talk sits near 0.94–0.97 while mean risk falls to the high 30s and low 40s, below the roughly 50 observed at cheap-talk values of 0.82–0.90. Rule S1 produces this pattern on maximally vague, uncommitted text because a paragraph making no claim is treated as a weak disclosure. The downturn is therefore an expected consequence of the locked rule base.

Discriminant validity is assessed within the vaguest quartile under the cheap-talk measure ($n = 253$). The baseline treats this group as nearly undifferentiated, with a cheap-talk mean of 0.96 and a standard deviation of 0.01. Within the same quartile, Figure 3 shows that GreenRisk scores span 35 to 90 (mean 38.6) and correlate with commitment at Spearman $\rho = 0.433$. The added signal separates “vague and quiet” text, a weak disclosure with nothing to evaluate, from “vague and loud” text, a confident, unsubstantiated pledge. This separation is the measured value that the four-signal design adds to the single-signal baseline.

A descriptive cut by TCFD disclosure category reproduces a pattern reported in the cherry-picking literature on climate disclosure. Metrics sections are concrete and low-risk (mean cheap-talk 0.204, mean risk 21.0). Strategy sections are more vague and higher-risk (0.651, 37.4), while risk and governance sections are more vague still (0.865/42.0 and 0.850/43.9, respectively). Firms are precise about

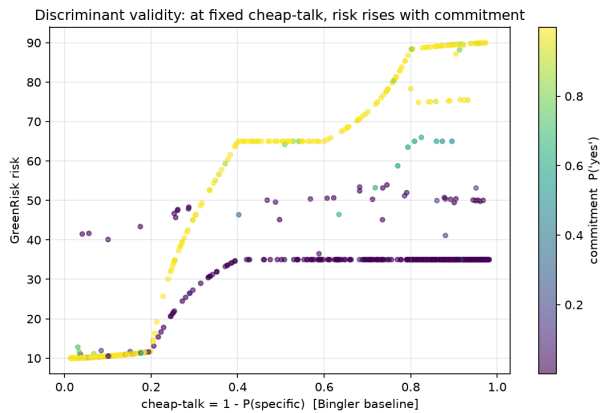


Figure 3: GreenRisk score distribution within the top cheap-talk quartile ($n = 253$), a group the baseline treats as essentially undifferentiated.

the numbers and vague about the surrounding narrative; GreenRisk recovers this asymmetry without using the category labels as inputs.

5.2 Layer 2: held-out face validity

The second layer uses fifteen paragraph-level cases drawn from public, regulator-adjudicated or third-party-verified sources. The set contains nine greenwashing cases from enforcement actions and rulings involving DWS Group, Volkswagen AG, and HSBC UK Bank plc, along with six rigorous reference disclosures from Microsoft and Ørsted A/S. The primary comparison restricts the analysis to the in-scope, vagueness-based greenwashing cases and the rigorous reference group. GreenRisk separates these groups with an AUC of 0.867 ($n = 5$ vs. 6; Mann-Whitney $U = 26$, one-sided $p = 0.026$). This large effect is presented as a demonstration, not a statistically definitive result, because a held-out, hand-curated, regulator-grounded case set cannot be scaled like a corpus run. Sensitivity within the in-scope group is 3/5 (0.60), and specificity within the reference group is 5/6 (0.83).

Figure 4 shows the score distribution across strata. All four of the out-of-scope, specificity-based greenwashing cases scored below the flag threshold of 50, as predicted before scoring. This result confirms that the instrument's blind spot is bounded and predictable.

Three traced cases illustrate what these numbers mean at the level of individual text. Case GW-003, DWS Group's "ESG is in our DNA" language, scores 89.7, High, through rule S3: low specificity paired with high commitment, the loud-pledge-no-substance signature the instrument is built to catch.

Case RD-006, an Ørsted disclosure describing its preparation for EU CSRD/ESRS reporting, scores 72.7, Elevated, through rule S6. This false positive arises from the vague-but-committed signature that flags genuine greenwashing elsewhere. The paragraph describes a reporting process in confident commitment language without citing figures, and the commitment classifier cannot distinguish that language from a hollow pledge.

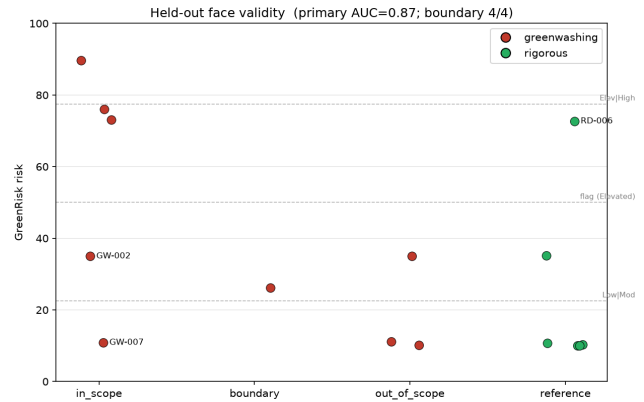


Figure 4: GreenRisk score by stratum in the held-out contrast set: in-scope greenwashing, boundary, out-of-scope greenwashing, and rigorous reference cases.

Case GW-008, HSBC's tree-planting advertisement, scores 10.17, Low, through rule S9 and falls outside the instrument's scope. The paragraph names concrete numbers and a defined program. Its omission—financed fossil-fuel emissions roughly fifty times larger than the advertised offset—is not visible to a classifier that reads only the surface form of the text.

6 Discussion

Section 5's numbers are best read alongside their limits, and this section makes those limits explicit.

6.1 The symmetric specificity-as-substance limitation

The central limitation is symmetric and follows from what the specificity signal measures. specificity reads whether a paragraph's surface form contains concrete figures, dates, or named mechanisms; it does not, and cannot, read whether those figures are honest, complete, or selectively chosen. Case RD-006 shows the low-specificity failure mode, a rigorous discloser using confident process language without numbers, scored as a false positive through the same signature that correctly flags genuine greenwashing. Cases GW-004 and GW-008 show the mirror failure mode, specific, numerically dense text that is nonetheless selective or corrupted in what it omits, scoring Low because it is specific. Put plainly, this work separates loud-vague from quiet-vague; it cannot separate specific-truthful from specific-but-selective, a distinction that would require verifying a paragraph's claims against external data, not just reading the paragraph itself.

6.2 The bounded commitment false-positive tail

A second, smaller limitation traces to the commitment signal specifically. Sized against 419 gated TCFD paragraphs, high commitment ($P \geq 0.90$) co-occurs with low specificity in 6 of 419 cases (1.4%). These cases are concentrated in the Elevated band, with one High-band case. The underlying classifier can fire strongly on generic policy or framework language, such as "our policy is..." even when

no substantive pledge is being made. The vagueness requirement in the DL-001 spine absorbs most of this over-confidence because specific text routes to the low-risk S8/S9 cells regardless of commitment strength. The residual 1.4% is bounded and documented. Correcting it would require intervention at the upstream classifier, not the rule level.

6.3 Sentiment collapse and Medium-band rarity

A further, deliberate simplification concerns how sentiment asymmetry is measured. The signal reads $P(\text{opportunity})$ alone and does not compute a signed balance of opportunity and risk framing. Risk-framed and neutral paragraphs consequently both collapse toward zero on this signal, so “balanced” framing does not sit at a distinguishable 0.5. The opportunity-framing tier (O1–O3) should therefore be read as detecting opportunity hype, not as a general sentiment axis. The spine cells also differ in empirical weight. Rule S5, which represents middling specificity and commitment, completes the full 3×3 tiling but rarely fires on disclosure text. Observations tend to cluster near the corners of the specificity-commitment space.

6.4 Scope of the evidence

External validity is bounded by corpus choice. Layer 1’s large-scale evidence comes entirely from TCFD-structured disclosure text, and TCFD’s own four-pillar structure, governance, strategy, risk management, and metrics and targets, may itself shape the kind of vagueness the instrument observes. The findings should not be generalized to CSRD-format disclosures, integrated reports, or marketing collateral without further testing. Layer 2’s AUC of 0.867 must likewise be read in light of the sample size. With five in-scope greenwashing cases and six reference cases, its confidence interval is wide. The one-sided p -value of 0.026 is below conventional significance thresholds, but it is not the strongest part of the claim. The evidence rests mainly on the large effect size, the AUC, and the use of regulator-adjudicated cases. A six-versus-five comparison is not a definitive statistical test.

6.5 Deployment framing

None of this evidence is intended to support autonomous flagging. GreenRisk is designed to function as decision support for a human auditor, analyst, or regulator, surfacing paragraphs and rule traces worth a closer look, not as a standalone verdict that a disclosure is or is not greenwashing. The activation trace allows a human reviewer to check, for any given score, whether the rule that fired matches their own reading of the paragraph, and can overrule the instrument when it does not.

7 Conclusion and Future Work

This paper tested whether greenwashing risk can be decomposed into a small set of construct-aligned signals and recombined into an auditable score. GreenRisk operationalizes this idea with four independently validated ClimateBERT signals and a locked, seventeen-rule Mamdani rule base. At corpus scale ($n = 1,009$), its score converges with the established cheap-talk baseline and still differentiates paragraphs within that baseline’s vaguest quartile. In the held-out case set ($n = 15$), the primary in-scope comparison yields

an AUC of 0.867. Within the limits established by the evaluation, these results support the narrower claim that a multi-signal rule base can add discrimination beyond vagueness alone while keeping the scoring path inspectable.

Future work should first quantify the contribution of each signal and rule tier through an ablation study that holds the rest of the rule base fixed. A second line of work should address the residual commitment false positives and the specificity-based false negatives identified in Section 6. Possible approaches include improving the upstream classifiers or adding a separate check for exclusion-list and policy-boilerplate text; either change would require a new instrument version. The broader research program will examine explainable, construct-aligned measurement in other domains where a single score must represent a multidimensional judgment.

Reproducibility Statement

Every result in this paper is reproducible against a fixed, versioned artifact. All five classifiers, the climate gate and the four risk signals, are loaded at pinned Hugging Face model revisions recorded in the project’s model registry, so that a later re-run reads the identical weights this paper reports on. The Python environment is pinned via uv, and every run of the pipeline emits a W3C PROV-O graph [4] binding its results to the exact model revisions, input data, and instrument version tag that produced them. The rule base and membership functions described in Section 3 are frozen under the tag `rulebase-locked-v1` (commit `a40288a`); any change after this point is logged as a separate instrument version. The instrument, its locked configuration, its evaluation artifacts, and this paper’s source are archived under an open Apache-2.0 license with a Zenodo DOI, 10.5281/zenodo.21122389, and maintained at <https://github.com/idcesares/GreenRisk>.

AI Usage Statement

Generative-AI tools were used as an execution and review layer during this work. Agent-based coding assistants supported the implementation, debugging, testing, and reproducibility engineering of the software pipeline. Language models accessed through APIs also assisted with organizing and analyzing a large body of candidate references and with reviewing the manuscript for grammar, structure, clarity, and internal consistency. All cited sources were checked by the authors against the original publications, and all AI-assisted code, analyses, and textual suggestions and writings were reviewed before inclusion.

The research question, domain analysis, construct selection, fuzzy-system design, evaluation protocol, interpretation of the evidence, and conclusions were determined by the authors. The AI systems did not act as autonomous researchers or authors and were not treated as sources of evidence. The authors retain full responsibility for the software, reported results, citations, and manuscript.

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Bingler et al. cheap-talk construct, and the scikit-fuzzy [7] implementation of Mamdani inference [5]; the acknowledgements page of the released artifact credits this work in full.

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